Homework 4

September 30, 2026

Work the following problems on engineering paper in the homework format as found on Canvas. All problems must be scanned and uploaded to Canvas by the assignment due date/time. Label each of your problems HW4 Problem X where X is 1, 2, or 3.

1. A binary channel (see Figure) is symmetric meaning that $\nu = \epsilon$. Assume the source X has a PMF given by

$$p_X(x) = \begin{cases} p & x = 1 \\ q = 1 - p & x = 0 \end{cases}$$

Answer the following questions.

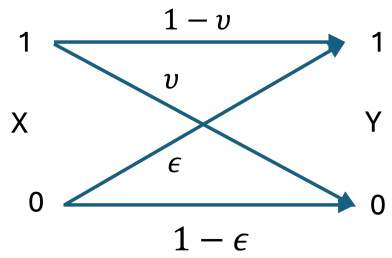


Figure 1: Binary channel for Problem 1.

   (a) Find formula for the entropy of the output $H[Y]$ and for the conditional entropy $H[Y/X]$ in terms of p and ϵ. If you replace ϵ with $1 - \epsilon$, do these formulas change?

   (b) Mutual Information $I[X, Y]$ measures the amount of information about X contained in Y and vice versa. It can be computed by

$$I[X, Y] = H[Y] - H[Y/X]$$

   If $\epsilon = 0.5$, find the mutual information. What does this mean?

2. The symbols $\{a, b, c, d, e, f\}$ have probabilities $\{0.23, 0.26, 0.04, 0.25, 0.17, 0.05\}$.

   (a) Find the entropy of the source.

   (b) Find the Huffman code for each letter and compute the expected length. *You must draw the tree to get credit for this problem.*

3. [1]The parity generator matrix for a Hamming (8,4) code is given by

$$\underline{\underline{G}} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

[1] You may use MATLAB for this problem if you like. If you do, provide a print out of your well commented code and a printout of any and all output the code produces. Label the top of each sheet as you would any problem done in the homework format.

(a) Compute the distance between all pairs of code words and show the distance of the code is 4. You may use MATLAB to do this.

(b) Show that the difference between any pair of code words is a code word.